Concept design

Core, fibre backbone, local PoE distribution, UPS protection and surveillance control room

MDF / CORE

LOCAL SERVEURS - ETAGE
Collapsed core/distribution, NVR/VMS servers, fibre patch panels and network security boundary.

CONTROL ROOM

LOCAL SECURITE - RDC BAS
Four-display operator position, VMS workstation/decoder and a local managed access switch.

LOCAL DISTRIBUTION

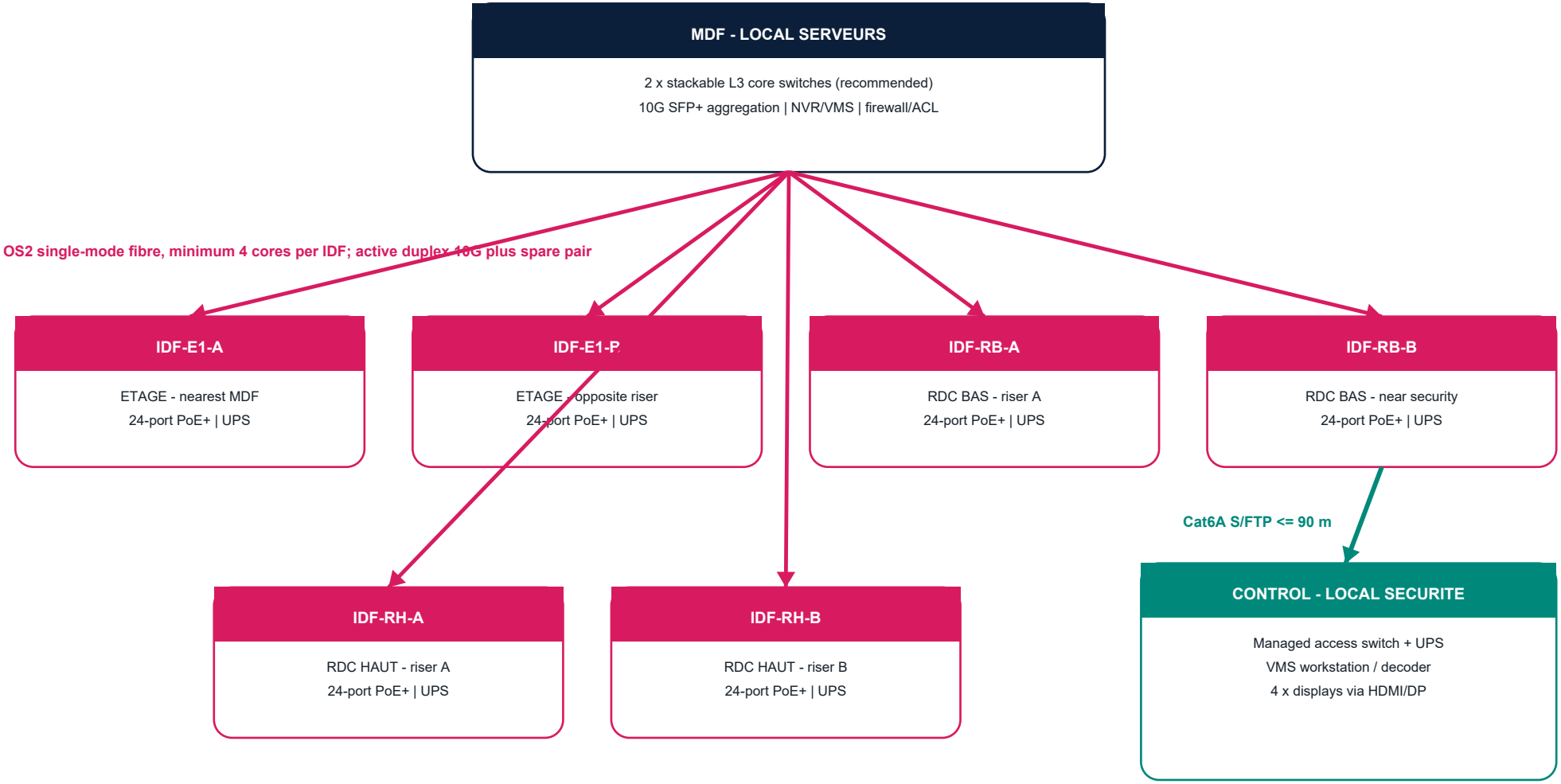
Six IDF nodes
Two CFA riser positions per level; each node receives one managed PoE+ switch and one local UPS.

DESIGN POSITION

The proposed architecture is sound if implemented as a star fibre backbone from every IDF to the MDF. Keep all CCTV equipment on the same physical infrastructure but segment it using dedicated VLANs and server-side ACLs. The switch in the control room is an operator/access switch, not a second independent core. The nearest RDC BAS IDF may feed it by Cat6A only after the permanent-link length is verified at 90 m or less. A 1000 VA / 600 W UPS is acceptable only where measured continuous load is 480 W or less; larger PoE loads or longer runtime require 1500 VA or greater.

Site constraint: The labelled CFA shafts are 1.00 x 0.20 m. Use an adjacent lockable 19-inch wall cabinet; do not place a normal switch and UPS inside a 200 mm-deep riser.

Logical topology



Monitors are display outputs from the VMS workstation/decoder, not direct camera network endpoints.

FAILURE-DOMAIN NOTE

A single core switch remains a single point of failure. Preferred implementation: two stackable/virtual-chassis core switches, dual power feeds where available, and two diverse fibre paths for critical IDFs. If budget permits only one core, install spare SFPs and keep a configured cold-spare switch.

Node and room schedule

Node	Drawing location	Role	Baseline equipment	Uplink / downstream
MDF-CORE	ETAGE - LOCAL SERVEURS	Core/distribution + recording	2 x L3 10G core switches; fibre ODF; NVR/VMS; dedicated UPS	Aggregates all IDFs
IDF-E1-A	ETAGE - CFA riser nearest LOCAL SERVEURS	Local PoE access	24-port managed PoE+; >= 2 x SFP+; UPS	10G OS2 to MDF
IDF-E1-B	ETAGE - opposite CFA riser	Local PoE access	24-port managed PoE+; >= 2 x SFP+; UPS	10G OS2 to MDF
IDF-RB-A	RDC BAS - CFA riser A	Local PoE access	24-port managed PoE+; >= 2 x SFP+; UPS	10G OS2 to MDF
IDF-RB-B	RDC BAS - CFA riser nearest LOCAL SECURITE	Local PoE + control-room feeder	24-port managed PoE+; >= 2 x SFP+; UPS	10G OS2 to MDF; Cat6A to CR
CR-SW	RDC BAS - LOCAL SECURITE	Operator/access switch	Managed switch; 1000-1500 VA UPS; VMS workstation/decoder	Cat6A from IDF-RB-B
IDF-RH-A	RDC HAUT - CFA riser A	Local PoE access	24-port managed PoE+; >= 2 x SFP+; UPS	10G OS2 to MDF
IDF-RH-B	RDC HAUT - CFA riser B	Local PoE access	24-port managed PoE+; >= 2 x SFP+; UPS	10G OS2 to MDF

CABINET RULE

Each IDF should be an adjacent lockable 6U-12U wall cabinet sized for the selected switch depth, fibre termination, copper patch panel, cable-management radius and UPS ventilation. Confirm clearances, door swing, service access, temperature and electrical earthing during the site survey.

Equipment, fibre and power specification

Component	Minimum concept requirement	Design check before purchase
IDF switch	Managed 24-port PoE+ (802.3at), VLAN/ACL, RSTP/MSTP, SNMP, >=2 x 10G SFP+	Camera count, per-camera watts, PoE budget and spare ports
Core	Two stackable/virtual-chassis L3 switches recommended; adequate 10G SFP+ density	Total uplinks, routing/ACL, multicast and NVR throughput
Fibre	OS2 single-mode, minimum 4 cores to each IDF, LC patching, LSZH/fire-rated cable	Measured route, fire stopping, bend radius and diverse route availability
Copper	Cat6A S/FTP permanent links; certified channel; max 90 m permanent link	Bonding/earthing and separation from power
Local UPS	1000 VA / 600 W minimum only when design load <=480 W	Actual watts, desired runtime, battery replacement and ambient temperature
Core UPS	Online/double-conversion preferred; capacity from measured core + server/NVR load	Generator interface, bypass, redundancy and runtime objective

POE BUDGET

For each IDF:
Required PoE budget >= sum(camera maximum watts) x 1.20
Then add the switch's own AC consumption when sizing the UPS. Reserve at least 20% switch ports and PoE budget for future devices. IEEE 802.3bt may be used where a future device needs four-pair higher-power PoE, but ordinary cameras usually fit PoE/PoE+.

UPS DECISION

Schneider Electric recommends keeping the connected load around 80% of UPS watt capacity. Therefore a 600 W unit should normally carry no more than about 480 W continuous. Runtime must be read from the selected model's load curve; 1000 VA does not guarantee a specific runtime. Use 1500 VA or higher if the load or autonomy target requires it.

BACKBONE MEDIA

OS2 is recommended for the building backbone because it supports long reach and future upgrades without replacing the cable. Use matched SFP+ optics at both ends, document optical loss, test every pair, and retain at least one spare duplex pair to each IDF.

Network segmentation and operations

Example segment	Example VLAN	Members	Permitted flows
CCTV cameras	120	IP cameras only	To VMS/NVR, DNS/NTP and approved management services; deny general Internet
VMS / operators	121	VMS servers, operator workstation, decoder	To cameras and recording services; controlled administrative access
Network management	199	Switch management, UPS network cards	From authorised admin hosts only; SNMPv3/syslog/NTP

ADDRESSING

Use fixed addressing or DHCP reservations for cameras and infrastructure. Suggested naming: CAM-[LEVEL]-[ZONE]-[NN], SW-IDF-[LEVEL]-[A/B], UPS-IDF-[LEVEL]-[A/B]. Maintain an IP/port/camera schedule and label both cable ends.

SECURITY

Unique camera credentials; disable unused services; HTTPS/SSH only; firmware lifecycle; no direct Internet exposure; role-based VMS accounts; central logging; configuration backups; restrict inter-VLAN access at the core.

RESILIENCE

Enable RSTP/MSTP or routed access as appropriate. Do not create an uncontrolled triangle between the MDF, nearest IDF and control-room switch. A second active path needs a documented STP/LACP/stacking design.

CONTROL ROOM SIGNAL FLOW

Cameras -> local PoE IDF -> 10G fibre -> MDF/core -> NVR/VMS. The control-room access switch reaches the VMS through IDF-RB-B. The four displays are driven from a VMS workstation or IP video-wall decoder by HDMI/DisplayPort. If the monitors are IP displays, confirm decoder bandwidth and vendor compatibility separately.

Before detailed design	Commissioning / acceptance
Count cameras per IDF and record maximum PoE watts	Certify every Cat6A permanent link and label both ends
Survey exact cabinet position beside each 200 mm CFA shaft	OTDR / insertion-loss test all fibre pairs; record results
Measure fibre and copper routes; confirm Cat6A <=90 m	Verify VLANs, ACLs, STP/LACP and management access
Confirm control-room furniture, sightlines, heat and four-display arrangement	Simulate switch/uplink failure and UPS transfer
Define recording retention, resolution, FPS and NVR storage	Verify camera time sync, recording, playback and export
Define required UPS autonomy and generator transfer time	Archive switch, UPS and VMS configuration backups

OPEN DECISIONS

1) Final camera count and wattage by zone. 2) Required recording retention and redundancy. 3) Exact rack locations adjacent to each CFA shaft. 4) Fibre route diversity. 5) UPS autonomy target. 6) Whether the core will be a redundant pair. 7) Whether the control room needs a second VMS workstation or only one workstation with four outputs.

Primary references

Cisco, Campus LAN and Wireless LAN Solution Design Guide - hierarchical access/distribution/core design and resiliency: https://www.cisco.com/c/en/us/td/docs/solutions/CVD/Campus/cisco-campus-lan-wlan-design-guide.html

IEEE 802.3-2022 / 802.3bt - Ethernet and four-pair Power over Ethernet: https://standards.ieee.org/ieee/802.3/10422/

Schneider Electric, Things to consider in sizing a UPS and load - headroom and runtime guidance: https://www.se.com/us/en/faqs/FAQ000268376/

Status: Concept design only. Final cable routing, fire-stopping, electrical protection, equipment selection and load calculations require an on-site survey and approval by the responsible low-current/electrical engineer.